

SEO Implementation Checklist for Orange Tree LMS (Next.js)

Overview

This document serves as the implementation checklist for Search Engine Optimization (SEO) in the Orange Tree LMS project. Complete each section before marking the project as production-ready.

Phase 1: Foundation

Domain & Security

- ☐ Configure production domain
- ☐ Enable HTTPS
- ☐ Configure SSL certificate
- ☐ Set preferred domain (www or non-www)
- ☐ Redirect HTTP → HTTPS

Analytics & Webmaster Tools

- ☐ Google Analytics 4
 - ☐ Google Search Console
 - ☐ Bing Webmaster Tools (Optional)
 - ☐ Verify website ownership
 - ☐ Submit sitemap.xml
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Phase 2: Technical SEO

Global Metadata

- ☐ Configure `metadataBase`
 - ☐ Add default title
 - ☐ Add title template
 - ☐ Add default description
 - ☐ Add keywords
 - ☐ Add favicon
 - ☐ Add theme color
 - ☐ Add language (`lang="en"`)
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Dynamic Metadata

Home Page

- ☐ SEO title
- ☐ Meta description
- ☐ Open Graph
- ☐ Twitter Card

Course Listing

- ☐ Dynamic metadata
- ☐ Category metadata

Course Details

- ☐ Dynamic title
- ☐ Dynamic description
- ☐ Open Graph image
- ☐ Canonical URL

Instructor Profile

- ☐ Dynamic metadata

Categories

- ☐ Dynamic metadata
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Phase 3: Robots & Sitemap

robots.txt

- ☐ Allow public pages
 - ☐ Block private pages
 - ☐ Add sitemap reference
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sitemap.xml

Include:

- ☐ Homepage
- ☐ Course listing
- ☐ Course pages

- ☐ Instructor pages
 - ☐ Category pages
 - ☐ Blog pages (if available)
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Phase 4: URL Structure

Public URLs

- ☐ `/`
- ☐ `/courses`
- ☐ `/courses/[slug]`
- ☐ `/categories/[slug]`
- ☐ `/instructors/[slug]`
- ☐ `/about`
- ☐ `/contact`

Avoid

- ☐ Query parameter URLs for public content
 - ☐ Duplicate URLs
 - ☐ Uppercase URLs
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Phase 5: On-Page SEO

Home Page

- ☐ One H1
 - ☐ Proper H2/H3 hierarchy
 - ☐ Internal links
 - ☐ Optimized content
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Course Details

- ☐ Course title
- ☐ Course description
- ☐ Learning outcomes
- ☐ Skills covered
- ☐ Prerequisites
- ☐ Instructor information
- ☐ Student reviews
- ☐ FAQ

- ☐ Related courses
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Instructor Profile

- ☐ Biography
 - ☐ Expertise
 - ☐ Published courses
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Phase 6: Structured Data (Schema.org)

Implement:

- ☐ Organization Schema
 - ☐ Website Schema
 - ☐ Course Schema
 - ☐ Breadcrumb Schema
 - ☐ FAQ Schema
 - ☐ Review Schema
 - ☐ Person Schema
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Phase 7: Image SEO

- ☐ Use Next.js Image component
 - ☐ Add descriptive alt text
 - ☐ Optimize image size
 - ☐ Lazy load images
 - ☐ Use WebP/AVIF where possible
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Phase 8: Performance

Core Web Vitals

Largest Contentful Paint (LCP)

- ☐ Under 2.5s

Interaction to Next Paint (INP)

- ☐ Under 200ms

Cumulative Layout Shift (CLS)

- ☐ Under 0.1
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Optimization

- ☐ Dynamic imports
 - ☐ Server Components where possible
 - ☐ Code splitting
 - ☐ Image optimization
 - ☐ Font optimization
 - ☐ Caching strategy
 - ☐ Compression enabled
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Phase 9: Accessibility

- ☐ Single H1 per page
 - ☐ Semantic HTML
 - ☐ Keyboard navigation
 - ☐ Proper focus states
 - ☐ Alt text
 - ☐ ARIA labels
 - ☐ Color contrast compliance
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Phase 10: Mobile SEO

- ☐ Fully responsive layout
 - ☐ No horizontal scrolling
 - ☐ Mobile-friendly navigation
 - ☐ Readable typography
 - ☐ Touch-friendly buttons
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Phase 11: Internal Linking

- ☐ Home → Categories
- ☐ Categories → Courses
- ☐ Courses → Related Courses
- ☐ Courses → Instructor
- ☐ Footer navigation

- ☐ Breadcrumb navigation
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Phase 12: Social Sharing

Open Graph

- ☐ Title
- ☐ Description
- ☐ Image
- ☐ URL

Twitter Cards

- ☐ Summary Large Image
 - ☐ Title
 - ☐ Description
 - ☐ Image
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Phase 13: Indexing Rules

Index

- ☐ Home
 - ☐ Course Catalog
 - ☐ Course Details
 - ☐ Categories
 - ☐ Instructor Profiles
 - ☐ About
 - ☐ Contact
 - ☐ Privacy Policy
 - ☐ Terms of Service
 - ☐ Blog (if public)
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No Index

- ☐ Login
- ☐ Register
- ☐ Forgot Password
- ☐ Dashboard
- ☐ Student Learning
- ☐ Instructor Dashboard

- ☐ Admin Dashboard
 - ☐ User Profile
 - ☐ Quiz Pages
 - ☐ Progress Pages
 - ☐ Certificates (unless intentionally public)
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Phase 14: Testing

SEO Validation

- ☐ Metadata validation
 - ☐ robots.txt validation
 - ☐ sitemap.xml validation
 - ☐ Structured data validation
 - ☐ Open Graph preview
 - ☐ Twitter Card preview
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Performance Testing

- ☐ Lighthouse Audit
 - ☐ PageSpeed Insights
 - ☐ Core Web Vitals
 - ☐ Mobile Friendly Test
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Search Console

- ☐ No crawl errors
 - ☐ No indexing errors
 - ☐ Sitemap processed successfully
 - ☐ Pages indexed correctly
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Implementation Priority

High Priority

- ☐ Global Metadata
- ☐ Dynamic Metadata
- ☐ Sitemap
- ☐ robots.txt

- ☐ Canonical URLs
 - ☐ Structured Data
 - ☐ Open Graph
 - ☐ Google Search Console
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Medium Priority

- ☐ Image Optimization
 - ☐ Performance Optimization
 - ☐ Accessibility
 - ☐ Internal Linking
 - ☐ Mobile Optimization
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Low Priority

- ☐ Twitter Cards
 - ☐ Bing Webmaster Tools
 - ☐ Additional Schema Types
 - ☐ Advanced Social Sharing
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Definition of Done

The SEO implementation is considered complete when:

- All public pages are indexable.
- All private pages are excluded from indexing.
- Every public page has unique metadata.
- Sitemap and robots.txt are accessible.
- Structured data validates successfully.
- Lighthouse SEO score is **95+**.
- Core Web Vitals meet Google's recommended thresholds.
- Google Search Console reports no critical SEO issues.